ITSG

100 programs, 800 million messages, 0 errors - AI-powered system review

About ITSG – who are we?

Innovative Technologies. Security. Together. This is what we stand for. Since 1996.

- **Why do we exist?**

We´re here to standardize and digitize the data exchange in healthcare and social insurance

- **Who are our customers?**

Exclusively our shareholders such as the National Association of Statutory Health Insurance Funds and their members.

- **Who are typical users?**

Anyone who exchanges data with the health insurance: employers and health service providers such as clinics or therapists.



About ITSG – who are we?

Four facts about ITSG

- Over 220 employees
- Over 46.5 million euros in sales in 2025
- Headquarters in Offenbach, branch in Berlin
- Four data centers of our own








ITSG GESAMTSYSTEM KRANKENVERSICHERTEN-NUMMER
ITSG TRUST CENTER
ITSG DAKOTA
Made in Germany !
ITSG PRIVATE PLATFORM
30 Jahre ITSG
GKV Datenschutz
BMG
BMAS
SYSTEM UNTER-SUCHUNG
GKV INFO SHOP
DATENSTELLEN
ITSG
SIEM / SOC
Informationssicherheit
RechenZentrum1
RechenZentrum2
Forum IT
Berlin
SV-MELDEPORTAL® ARBEITGEBER SOZIALVERSICHERUNG
Software-Entwicklung
Flexible Datendrehscheibe
Verzeichnisse und Basisdaten
Statistik-Küche
ITSG GKVNET CRYPT
INFORMATIONSPORTAL ARBEITGEBER SOZIALVERSICHERUNG
Frankfurt/Main
Atelier RAP
GKV-KOMSERVER
Müller & Co.KG
Offenbach
ITSG
Daten-Auto-bahn
Pflegeheim
Fachkonzept-Schmiede
KOHLE
MONZE
Strategie & Standards
Schütte 2026

About the ITSG – what do we do?

SV-MELDEPORTAL®
ARBEITGEBER
SOZIALVERSICHERUNG



SYSTEM
UNTERSUCHUNG

ITSG PRIVATE
PLATFORM

ITSG TRUST CENTER

System review – what is it?

- Systems = payroll programs (SAP, Datev, Lexware etc.)
- €400 billion in social insurance contributions
- 800 million messages exchanged per year
  - proof of contributions
  - sick notes
  - registration and deregistration
  - income replacement benefits
  - …



System review – what is it?

- Objective: To ensure that social insurance contributions are calculated correctly and messages are transmitted correctly.
- Specifications spread across 100 documents.
- Before the initial approval of a system, we provide intensive consultation and testing of the entire software.
- This is followed by an annual quality control (audit).



Our legal mandate

Book IV of the Code of Social Law (SGB) - Common provisions for social insurance - (Article I of the Act of 23 December 1976, Federal Law Gazette I p. 3845) - Section 95b System Review

(1) Persons subject to reporting obligations shall submit reports and proof of contributions by means of data transmission from system-tested programs or system-tested electronic filling aids. This shall also apply to applications and certificates, insofar as this is regulated under this Code or the Expenditure Compensation Act. . . .

(2) A system review shall be carried out for programs and electronic filling aids used for the exchange of data between persons subject to reporting obligations and the social security institutions and other accepting bodies pursuant to subsection (1). The system review shall include advice as well as the professional and technical examination of the application software for the recording, testing, administration, calculation and processing as well as the transmission, acceptance or retrieval of the necessary data. Payroll accounting programs shall carry out the calculations and the generation of data as well as their verification by machine; Filling aids shall support manual calculations by electronically transmitting and storing the data. . . .

(3) The hardware belonging to the information technology infrastructure of a reporting party, the operating systems and the internal communication software are not part of the system audit.

(4) The system review shall be carried out by the National Association of Health Insurance Funds with the participation of the pension insurance institutions, the accident insurance institutions and the Federal Employment Agency on behalf of all the umbrella organisations of the social insurance institutions and the Working Group of Professional Pension Institutions.

. . .

Source: § 95b SGB 4 - paragraph 2

The Challenge

Design an AI solution that supports and potentially even replaces the testing processes.

For this challenge, **focus on the payroll process for income replacement benefits** ("Entgeltersatzleistungen")
e.g. maternity benefit and sickness benefit.

It is intended to automatically check whether a program meets all applicable specifications within this scope.

The AI solution is intended to offer different technical approaches.



How it could work

1. AI generates technical test cases.
2. Humans perform test cases in payroll software.
3. AI checks the data sets generated in the process.

Optional: The **AI review assistant** is set up locally by the software vendor who then tests the application. The AI review assistant generates a test report for the ITSG system review.



Objectives: What should the AI review assistant be able to do?

- **Identify** the relevant requirements from the specifications
- **Translate** the requirements into testable criteria
- **Generate** test cases with expected results
- **Create** a traceable test record
- **Test** the payroll software with the test cases **and compare** the results with the expected results
- **Deliver** comprehensible test reports

The cherry on top

As an optional extension, build a prototype based on your concept.

The prototype can be implemented as a code demonstrator, app, website or in another suitable form, provided that it is clearly recognizable how the solution supports the system review process.



If you have any questions, we are happy to help you



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